

SUKRITH POTHNAK

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience designing, deploying, and scaling production-grade machine learning and Generative AI systems. Proven expertise in building end-to-end AI pipelines using LLMs, RAG architectures, and MLOps frameworks with strong focus on scalability, latency optimization, and business impact. Experienced in delivering AI solutions on **AWS, Azure, and GCP** using modern AI stacks including **LangChain, LlamaIndex, OpenAI APIs, PyTorch, TensorFlow, MLflow, Kubeflow, and Kubernetes**. Strong background in **Data Engineering (Spark, Kafka, Databricks), API development (FastAPI), and cloud-native AI deployment**. Passionate about building **production-ready AI systems, AI agents, and scalable ML platforms** that bridge research and real-world applications.

TECHNICAL SKILLS

Programming: Python, SQL, Scala, Bash
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow
Deep Learning: Transformers, CNN, LSTM, Diffusion Models, Fine-tuning (LoRA, PEFT)
GenAI / LLM Stack: OpenAI API, Claude API, HuggingFace, LangChain, LlamaIndex, RAG, Prompt Engineering
Vector Databases: Pinecone, Weaviate, FAISS, ChromaDB, Milvus
MLOps: MLflow, Kubeflow, DVC, Airflow, Model Registry, CI/CD (GitHub Actions)
Cloud Platforms: AWS (SageMaker, Lambda, EKS), Azure ML, GCP Vertex AI
Big Data: Spark, Kafka, Databricks, Hadoop
APIs & Backend: FastAPI, Flask, REST APIs, Microservices
DevOps: Docker, Kubernetes, Terraform, Helm
Monitoring & Observability: Prometheus, Grafana, Evidently AI, Drift Detection
Data Tools: Pandas, NumPy, SQL, Snowflake, BigQuery
Visualization: Power BI, Tableau, Plotly

PROFESSIONAL EXPERIENCE

AI/ML Engineer Meta, USA	May 2024 – Present
- Designed and deployed enterprise-scale Generative AI and LLM solutions using GPT, Llama, and Claude models with RAG architectures, enabling intelligent knowledge retrieval and content generation for Meta platforms. - Built and optimized AI Agent frameworks using LangChain, LlamaIndex, and multi-agent orchestration techniques, automating workflows and improving user engagement across internal AI applications. - Developed vector search and semantic retrieval systems using Pinecone and FAISS, improving search relevance and retrieval accuracy by 35% across large-scale knowledge repositories. - Engineered end-to-end MLOps pipelines using Kubeflow, MLflow, and CI/CD automation, reducing model deployment cycles by 70% and accelerating experimentation. - Optimized LLM inference performance through model quantization, GPU acceleration, and distributed serving, reducing latency by 40% while maintaining model quality. - Built scalable AI-powered microservices using FastAPI, Docker, Kubernetes (EKS), and AWS services, supporting 1M+ daily inference requests with high availability. - Implemented real-time model monitoring, drift detection, and observability frameworks using Evidently AI, Prometheus, and Grafana, improving production reliability and governance. - Collaborated with Data Scientists, Product Managers, and Software Engineers to integrate AI capabilities into Meta's recommendation, search, and content intelligence ecosystems. - Developed and fine-tuned transformer-based models for personalization, ranking, and content understanding, improving user experience metrics by 25%. - Leveraged distributed data processing frameworks and cloud-native architectures to train and deploy AI models on petabyte-scale datasets.	
Impact:	
- Reduced AI model deployment time from weeks to hours through automated MLOps pipelines. - Improved semantic search relevance and retrieval accuracy by 35%.	

- Reduced inference latency by 40% through optimization and scalable serving strategies.
- Enabled reliable processing of 1M+ AI requests per day in production environments.
- Increased AI-driven user engagement and recommendation effectiveness by 25% across targeted applications.

Data Engineer | Amazon, India

Jan 2021– June 2023

- Developed and maintained data pipelines for processing large-scale e-commerce datasets (millions of daily records), supporting analytics and early ML use cases.
- Worked with Apache Spark (PySpark) and SQL to perform data transformation and aggregation, improving query efficiency by 20% for downstream analytics.
- Built and monitored batch workflows using Apache Airflow across 15+ pipelines, ensuring reliable data availability and 95%+ SLA adherence.
- Collaborated with data science teams to support feature engineering and data preparation, contributing to improved model input quality and consistency.
- Utilized AWS (S3, EC2, Lambda) for scalable data storage and processing, handling high-volume structured and semi-structured data.
- Performed data cleaning and validation using Python (Pandas) and SQL, reducing data inconsistencies by 10–15% in reporting and ML datasets.
- Optimized SQL queries and Spark jobs, reducing batch processing time by 20% for key workflows.
- Documented data pipelines and contributed to standardization efforts, helping reduce onboarding time for new team members by 15–20%.

ACADEMIC PROJECTS

Intelligent RAG-Based Research Assistant (GenAI + LLMs)

Overview: Developed an AI-powered research assistant using Retrieval-Augmented Generation (RAG) to answer domain-specific queries from large document collections such as research papers and PDFs.

- Built end-to-end pipeline using LangChain + OpenAI API + HuggingFace embeddings.
- Implemented semantic search using FAISS vector database for high-relevance retrieval.
- Designed context-aware prompting to improve LLM response accuracy and reduce hallucinations.
- Deployed backend using FastAPI, enabling real-time query processing.
- Improved answer relevance by 35% compared to traditional keyword search.

Real-Time ML-Based Customer Churn Prediction System

Overview: Developed a scalable machine learning system to predict customer churn for subscription-based services using real-world-like behavioral datasets.

- Built classification models using XGBoost, LightGBM, and Scikit-learn.
- Performed advanced feature engineering and data preprocessing on large datasets.
- Implemented end-to-end ML pipeline with Airflow + MLflow for tracking and orchestration.
- Deployed model using AWS SageMaker + REST API (Flask) for real-time predictions.
- Achieved 92% prediction accuracy with optimized recall for churn cases.

EDUCATION:

Master's Degree in Data Science | Florida State University, Tallahassee

Aug 2023 – May 2025

CERTIFICATION:

Microsoft Certified: Azure Data Scientist Associate

AWS Certified Cloud Practitioner